

ClaimSense 360: Enterprise AI Insurance Claims Intelligence Platform

MASTER TECHNICAL ARCHITECTURE, AI/ML SPECIFICATIONS & DEFENSE MANUAL

Executive Overview: ClaimSense 360 is an Enterprise & Government-Grade AI Motor Insurance Claims Intelligence and Anti-Fraud Decisioning Platform. It automates end-to-end evaluation using a multi-modal artificial intelligence engine combining Tabular Machine Learning (XGBoost), Game-Theoretic Explainability (SHAP), Natural Language Processing Deception Detection (TF-IDF NLP), and Deep Learning Computer Vision (Ultralytics YOLOv8 + PyTorch ResNet-18) alongside Digital Image Forensics (EXIF Anti-Spoofing).

1. Industry Problem & Executive Rationale

- The ₹45,000 Crore Annual Loss:** In India alone, motor insurance fraud causes losses exceeding ₹45,000 Crore (\$5.4 Billion USD) annually due to staged collisions, inflated repair estimates, and recycled damage images.
- Failure of Legacy Rule-Based Systems:** Traditional insurance platforms rely on static IF-THEN rules (e.g., 'If claim > ₹1 Lakh, send to surveyor'), which fraud syndicates easily bypass.
- The Black-Box AI Problem Solved:** Regulators (IRDAI/DPDP) reject black-box AI scores. ClaimSense 360 solves this by offering mathematical feature-level SHAP explainability and EXIF anti-spoofing telemetry.

2. Full-Stack Architecture & Technology Stack

ClaimSense 360 uses a modern decoupled microservices architecture:

Layer	Technology Used	Role & Function
Frontend Portal	Next.js 15.1 + React 19 + TypeScript + Tailwind CSS	Sub-second responsive App Router UI, type-safe state, Recharts analytics.
Backend Service	Python 3.13 + FastAPI 0.110 + Uvicorn ASGI	Async REST microservice, SQLite DB (SQLAlchemy ORM), JWT Bearer auth.
Tabular ML	XGBoost 2.0 + SHAP 0.44 TreeExplainer	Claim fraud probability scoring + game-theoretic marginal contribution.
NLP Engine	TF-IDF Vectorizer + Ridge Regression	Incident description deception detection & 1-gram/2-gram phrase extraction.
Computer Vision	Ultralytics YOLOv8 + PyTorch ResNet-18	22-class vehicle damage object detection + L2 deep feature vector norm.
Forensics	PIL EXIF Telemetry Inspector	Scans raw camera headers to catch web/Google downloaded stock photos.

3. AI/ML Engine Models & Mathematical Foundations

- 3.1 XGBoost Fraud Classification Engine:** Trained on 1,000 real motor insurance claims across 46 financial and demographic features.
- 3.2 SHAP Explainability Formula:** Calculates exact marginal feature contribution using game theory:

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \left[\frac{|S|!(|N|-|S|-1)!}{|N|!} \right] * [v(S \cup \{i\}) - v(S)]$$

3.3 Computer Vision Damage Severity Formula: Combined YOLOv8 + ResNet-18 score:

Severity Score = 0.50 * S_YOLO + 30 * C_ResNet + 20 * E_Gradient

3.4 Digital Image Forensics & EXIF Anti-Spoofing: Detects missing smartphone camera headers on web-downloaded images, automatically adding +1.850 to SHAP risk and flagging High Risk for SIU audit.

4. Datasets Integrated

- 4.1 Kaggle Insurance Claims Real Dataset: 1,000 real motor insurance claim records (located at backend/app/data/insurance_claims_real.csv). Seeded 15 authentic Kaggle policy records into SQLite DB with natural customer names mapped to real policy IDs (POL-521585, POL-342868).
- 4.2 Roboflow Universe Car Damage Dataset: 32,664 annotated vehicle damage images across 22 damage classes (front-bumper-dent, headlight-shatter, side-fender-crush, etc.).

5. Page-by-Page Portal Overview

1. Executive Dashboard (<http://localhost:3000>): 4 Live KPI Stat Cards, Interactive Risk Simulator, Portfolio Risk Chart.
2. Claims Directory (<http://localhost:3000/claims>): Live search by customer name, policy number, or vehicle model with direct View Detail actions.
3. Single Claim 360 Inspector ([http://localhost:3000/claims/\[id\]](http://localhost:3000/claims/[id])): Policy Info Card, Incident Details Card, Uploaded Damage Photo Evidence Card, SHAP Math Factor Bars, and EXIF Forensics Warning Badge.
4. Claim Intake Form (<http://localhost:3000/claims/new>): Form presets for Nihar Sahu (Hyundai Creta) & Shubhansu Sahu (Suzuki Access), image upload dropzone, and sub-second execution.
5. SIU Fraud Queue (<http://localhost:3000/fraud>): Interactive Severity Filter Tabs (All Flagged, Critical Risk >= 78, Major Risk 50-77).
6. Portfolio Analytics (<http://localhost:3000/analytics>): Loss ratio bar charts and monthly risk trends.
7. AI Copilot (<http://localhost:3000/copilot>): Conversational AI claims intelligence assistant.

6. System Hardware Specifications

Component	Minimum Hardware	Recommended / Actual Test Specs
CPU Processor	Intel i5 (4 Cores) or AMD Ryzen 5	AMD Ryzen 7 8845HS / 7000 Series (8 Cores / 16 Threads)
RAM Memory	8 GB DDR4	16 GB DDR5
GPU Graphics	Integrated Intel UHD / AMD Radeon	NVIDIA GeForce RTX 4060 Laptop GPU (8GB VRAM, CUDA 12+)
Storage Space	5 GB available SSD space	NVMe M.2 High-Speed SSD
Operating System	Windows 10 / 11 64-bit, macOS, Linux	Windows 11 Home 64-bit
Runtimes	Node.js v18+, Python 3.10+	Node.js v20+, Python 3.13

7. How to Start and Stop the Application

To Start Next.js Frontend Server (Port 3000):

```
cd "c:\Users\NIHAR\Documents\AD Research\CLAUDE AI\files\ANTIGRAVITY\claimsense360" npm run dev
```

To Start Python FastAPI Backend Server (Port 8000):

```
cd "c:\Users\NIHAR\Documents\AD Research\CLAUDE AI\files\ANTIGRAVITY\claimsense360\backend"  
python -m uvicorn app.main:app --host 127.0.0.1 --port 8000
```

To Stop the Servers: Press CTRL + C in terminals, or force terminate port 8000 via PowerShell using: `taskkill /F /PID`